Industry Best Practices: Public IP Addresses and Network Connectivity

Optimize network cost only after establishing the workload's connectivity, security, routing, availability, and dependency requirements. Network resources often support multiple applications and should be reviewed as shared infrastructure.

Applicable resource types

Microsoft.Network/publicIPAddresses and related network interfaces, load balancers, gateways, firewalls, private endpoints, and routing resources.

Cost optimization practices

Maintain a current inventory of address allocation method, association, SKU, zone requirements, DNS records, firewall rules, routing dependencies, and owning workload. Review unassociated addresses as candidates for an ownership and dependency assessment.

• Use the least costly architecture that still meets availability, security, routing, and IPv4/IPv6 requirements.

• Monitor data-transfer patterns and cross-region traffic before redesigning connectivity.

• Tag network resources with owner, environment, application, cost center, and planned end date where applicable.

• Use Cost Management and Azure Advisor signals as triage inputs, then validate them with the network and application owners.

Operational safeguards

An unassociated public IP may still be reserved for failover, migration, allowlists, DNS cutover, or an approved upcoming deployment.

• Require dependency and change review before altering an allocation.

• Confirm DNS, firewall, load-balancer, API allowlist, and disaster-recovery implications.

• Document an approved rollback and communication plan for customer-facing endpoints.

Evidence and approval model

Decision area	Evidence to collect	Safe action gate
Unassociated allocation	Association state, owner, DNS/allowlists, planned change, disaster-recovery role	Network-owner dependency review and approved change record
Network redesign	Flow logs, egress pattern, latency, resilience and security architecture	Architecture review; validate resilience and cost trade-offs
Cost attribution	Tags, subscription/resource-group scope, data-transfer billing dimensions	FinOps owner validates allocation completeness

Source material

This reference summarizes the following official guidance. Check the linked sources for current service limits, billing terms, and implementation details.

- **Azure Cost Management recommendation workflow**
https://learn.microsoft.com/en-us/azure/cost-management-billing/costs/tutorial-acm-opt-recommendations

- **Azure Well-Architected Framework overview**
https://learn.microsoft.com/en-us/azure/well-architected/what-is-well-architected-framework

- **FinOps Framework overview**
https://learn.microsoft.com/en-us/cloud-computing/finops/framework/finops-framework